background, which allow a separate test of this conclusion, verify the result: Dark matter and nuclear fusion don't mix.

Thus, as best we can figure, the dark matter doesn't simply consist of matter that happens to be dark. Instead, it's something else altogether. Dark matter exerts gravity according to the same rules that ordinary matter follows, but it does little else that might allow us to detect it. Of course, we are hamstrung in this analysis by not knowing what the dark matter is in the first place. If all mass has gravity, does all gravity have mass? We don't know. Maybe there's nothing wrong with the matter, and it's the gravity we don't understand.

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The discrepancy between dark and ordinary matter varies significantly from one astrophysical environment to another, but it becomes most pronounced for large entities such as galaxies and galaxy clusters. For the smallest objects, such as moons and planets, no discrepancy exists. Earth's surface gravity, for example, can be explained entirely by the stuff that's under our feet. If you are overweight on Earth, don't blame dark matter. Dark matter also has no bearing on the Moon's orbit around Earth, nor on the movements of the planets around the Sun—but as we've already seen, we do need it to explain the motions of stars around the center of the galaxy.

Does a different kind of gravitational physics operate on the galactic scale? Probably not. More likely, dark matter consists of matter whose nature we have yet to divine, and which gathers more diffusely than ordinary matter does. Otherwise, we would detect the gravity of concentrated chunks of dark matter dotting the universe—dark matter comets, dark matter planets, dark matter galaxies. As far as we can tell, that's not the way things are.

What we know is that the matter we have come to love in the universe—the stuff of stars, planets, and life—is only a light frosting on the cosmic cake, modest buoys afloat in a vast cosmic ocean of something that looks like nothing.

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During the first half million years after the big bang, a mere eyeblink in the fourteen-billion-year sweep of cosmic history, matter in the universe had already begun to coalesce into the blobs that would become